Assignment

1. Explain publicly accessible MIMIC data set. (7 Marks)
2. Identify types of healthcare datasets. (2 Marks)
3. Explain healthcare data sources and preprocessing steps. (7 Marks)
4. Given the MIMIC-IV patients, admissions and icustays tables:

    a) Identify the primary identifier in each table.
    b) Identify `subject_id`, `hadm_id` and `stay_id`.
    c) Explain the relationship between the three tables.
    d) Identify any three demographic/admission attributes from the table

5. Consider the MIMIC-IV chartevents and d_items tables.

A chartevents record contains:

subject_id = 10001

hadm_id = 20001

stay_id = 30001

itemid = 220045

value = 88

valueuom = bpm

The corresponding d_items record identifies itemid 220045 as Heart Rate.

a) What does subject_id represent?

b) What does stay_id represent?

c) What is the purpose of itemid?

d) Why is d_items required to interpret chartevents?

e) What clinical observation is represented by the record?

f) What is its recorded value and unit?

6. Using the MIMIC-IV diagnoses_icd and d_icd_diagnoses tables:

A patient has the following icd_code values:

4139

41401

25002

V8532

a) Identify the table containing the diagnosis codes.

b) Identify the dimension table used to obtain their descriptions.

c) Identify the attribute used to link the two tables.

d) Explain why the ICD code alone is insufficient for a human-readable interpretation.

e) Decode the diagnosis codes using the dimension table.